

```
def set_mesh_information(_):  
    g = create_grid  
    Z = micro_topography  
    d = {diffusion}  
  
def update_until(t, data):  
    A_v = data["z velocity"]  
    timestep = t - current_time  
    dt = timestep / substeps  
    dz = 0  
    for _ in range(substeps):  
        z0 = Z.copy()  
        Z += A_v * dt  
        Z = diffusion(e)  
        dz += Z - z0  
    current_time = t  
  
    return dz
```